Predictive and Generative Modeling of Therapeutic Antibody Developability Using Fine-Tuned Protein Language Models

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Abstract

Therapeutic antibody development is a costly and time-intensive process, with many candidates failing due to poor developability [1]. Developability metrics are defined by biophysical properties such as thermostability, self-association propensity, hydrophobicity, and solubility [1]. In this project, we investigate whether sequence-based protein language models can learn representations that (A) predict developability metrics and downstream therapeutic antibody approval, as well as (B) enable lead optimization of developability properties through guided sequence generation.

We have implemented two complementary baselines. For approval prediction (Task A), we trained a bootstrapped XGBoost classifier [2] on 11 experimentally measured biophysical features from the Ginkgo Bioworks PROPHET-Ab dataset [3], achieving a median AUPRC of 0.665 across 20 stratified splits. For developability metric prediction, we trained Ridge regression models on p-IgGen [4] sequence embeddings and evaluated them under a hierarchical cluster cross-validation scheme. Spearman correlations varied substantially across metrics under 5-fold cross-validation, with the strongest signals for PR_Ova ($\rho = 0.549$), PR_CHO ($\rho = 0.496$), and AC-SINS_pH7.4 ($\rho = 0.394$); HIC achieved $\rho = 0.335$, while T_{m2} and SEC %Monomer were near zero or negative.

In parallel, we have implemented a generative developability optimization pipeline for Task B. The Ginkgo Bioworks PROPHET-Ab dataset [3] was split into 5 folds using hierarchical-cluster, IgG isotype-stratified splits. The pipeline trains a FiLM (Feature-wise Linear Modulation)-conditioned velocity network using the conditional flow matching (CFM) objective in a 10-dimensional PLS-reduced CDR3 embedding space, guided by a Ridge regression oracle trained on full-sequence AbLang2 embeddings. The baseline oracle achieves Spearman $\rho = 0.321$ for HIC and $\rho = 0.445$ for AC-SINS on the held-out fold-1 test set. The unguided baseline generates CDR3 regions using AbLang2 decoding conditioned only on the parent framework, while the guided condition additionally uses the flow model to steer generation toward improved developability targets. Evaluated across 50 held-out test antibodies, both the flow-guided and unguided AbLang2 generation conditions achieve statistically significant improvements over parent antibodies. The unguided model achieved the following performance: oracle HIC ($\Delta = -0.271$, $p < 0.001$), oracle AC-SINS ($\Delta = -4.29$, $p < 0.001$), CamSol solubility ($\Delta = +0.57$, $p < 0.001$), and SAP hydrophobicity score ($\Delta = -0.23$, $p < 0.001$). The guided model showed comparable performance: oracle HIC

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($\Delta = -0.245$, $p < 0.001$), oracle AC-SINS ($\Delta = -4.18$, $p < 0.001$), CamSol solubility ($\Delta = +0.653$, $p < 0.001$), and SAP hydrophobicity score ($\Delta = -0.207$, $p < 0.001$). Guided and unguided generation perform comparably, suggesting that AbLang2's language model prior is the primary driver of improvement at this time point.

The second half of the project will focus on fine-tuning AbLang2 [5] for improved sequence-based prediction, conditioning the lead optimization to maintain antigen binding affinity and specificity, pretraining the flow model on the OAS dataset [6] to address data scarcity and framework-dominance issues, and conducting a full comparative evaluation across prediction and generation tasks.

1 Introduction and Project Overview

Therapeutic antibodies are among the most widely used classes of protein-based biotherapeutics, but their development remains expensive and time-consuming [7] [8]. A central bottleneck is lead optimization, in which one candidate is selected from a small set of molecules with similar bioactivity based on predicted manufacturability, stability, and other biophysical properties [1]. This selection is based on intrinsic biophysical properties such as thermostability, self-association propensity, hydrophobicity, and solubility [9]. These properties determine whether a candidate can be successfully manufactured and whether it will remain stable in clinical settings [9]. Identifying and correcting poor developability early in the design process (before moving candidates forward in costly clinical trials) is therefore an unmet need in the discovery of antibody therapeutics [9]. Ideally, such tools should operate with input only from the amino acid sequence, without requiring expensive experimental assays [10].

Therefore, our refined research question is as follows: Can sequence-based antibody protein language models (A) predict developability metrics and downstream therapeutic approval, as well as (B) enable optimization of true developability properties through guided sequence generation? This research question explicitly separates two distinct tasks: direct regression and prediction of biophysical developability metrics (HIC, AC-SINS, T_m, etc.) and optimization of true developability properties from guided sequence generation using diffusion-based approaches. With respect to the original project proposal, we have updated our modeling approach such that our final model consists of an AbLang2 encoder [5] fine-tuned on the GDPa1 dataset. We will compare the fine-tuned AbLang2 performance against baseline ridge regression models trained on p-IgGen embeddings [4]. We also substantially redesigned the generative pipeline to operate in a PLS-reduced CDR3 embedding space with FiLM conditioning [11] rather than iterative branch-and-score integration.

Our current contributions at the midterm are:

- An XGBoost baseline for approval prediction trained on experimentally measured biophysical features, with bootstrapped performance estimates (median AUPRC 0.665).
- Ridge regression baselines for developability metric prediction trained on p-IgGen embeddings and evaluated under stratified 5-fold cross-validation. These regression models can also serve as the scoring oracle for guiding sequence generation in Task B.
- An implemented conditional flow matching pipeline for CDR3 optimization, with preliminary quantitative results demonstrating significant improvements in AC-SINS, CamSol, and SAP relative to parent antibodies.
- Identification of two key challenges (data scarcity and framework-embedding dominance) with mitigation strategies planned for the second half of the project.

2 Background (Brief)

2.1 Antibody Developability and the Need for Sequence-only Prediction

Developability refers to the set of biophysical properties an antibody candidate must satisfy to be successfully manufactured and safely administered [1]. The most clinically relevant metrics are thermal stability (T_{m1}, T_{m2}), hydrophobicity and non-specific surface binding (HIC, SMAC, HAC,

SAP score), self-association and polyreactivity (AC-SINS, PR_CHO, PR_Ova), and colloidal stability (SEC %Monomer, Expression Titer) [1][3]. Poor values in any of these areas can cause aggregation, manufacturing failure, or adverse immune responses. Identifying developability liabilities early (before expensive clinical trials) is a key unmet need in antibody discovery [9]. A critical limitation of prior predictive work (Arsiwala et al.) is that it relies on experimentally measured biophysical features as inputs [3]. One practical limitation of this strategy is that predicting approval requires access to properties that are typically available only after experimental characterization. Another challenge is that developability datasets are typically small in size (containing hundreds to low thousands of antibodies) compared to the vast sequence space of potential antibody candidates [3]. The work encompassed in this project focuses on sequence-only prediction, which is more relevant for early-stage lead evaluation and optimization.

2.2 Protein Language Models and Antibody Language Models

Antibody language models pretrained on large sequence corpora can encode structural and biophysical information without explicit physics-based calculations [10]. Linear probing of pretrained embeddings with Ridge regression is a strong and interpretable baseline for developability prediction, with Spearman correlations of 0.3–0.7 reported for thermostability on small datasets [12]. However, general-purpose protein language models such as ESM2 (previously used for antibody developability prediction and CDR design) exhibit a well-documented germline bias [5]. Due to its pretraining bias toward germline-proximal sequences, ESM2 assigns low likelihood to somatic hypermutations in CDR regions, which are essential for antigen specificity and drive developability differences in therapeutic antibodies [5]. AbLang2 was designed to correct for this bias by downweighting germline sequences during pretraining, producing embeddings that more faithfully represent the somatic diversity observed in therapeutic antibodies [5]. AbLang2 processes paired heavy- and light-chain sequences using a bidirectional transformer, capturing germline gene usage, CDR loop length distributions, and residue-level co-evolutionary patterns. For task-specific adaptation, we plan to apply LoRA fine-tuning [13] to avoid overfitting on the PROPHET-Ab dataset.

2.3 Generative Models for Antibody CDR Optimization.

A compelling recent approach from Ginkgo Bioworks trains a discrete diffusion model on 1.68 million paired OAS sequences (using ESM2 embeddings as the sequence representation) to generate developable CDR3 variants [14]. This pipeline inherits ESM2's germline bias in its backbone embeddings, potentially limiting its ability to model the somatic mutations most relevant to developability optimization. We instead use AbLang2 as the backbone, providing embeddings better suited to the therapeutic antibody sequence distribution. Beyond the choice of backbone, we replace discrete diffusion with continuous conditional flow matching [15]. Flow matching trains a velocity network to transport Gaussian noise to the data distribution along straight-line interpolant trajectories, rather than the curved, noise-schedule-dependent paths of score-based diffusion. This yields faster training convergence, more stable ODE integration at inference, and straightforward FiLM-based conditioning [11] on continuous developability target values.

3 Data

3.1 PROPHET-Ab (GDPa1)

Our primary dataset is Ginkgo Bioworks PROPHET-Ab [3], which is a publicly available high-throughput platform dataset containing 246 therapeutic antibodies with experimentally measured biophysical properties and paired variable heavy (VH) and variable light (VL) chain amino acid sequences. The antibodies included in PROPHET-AB are 106 FDA-approved, 135 clinical-stage, and 5 withdrawn therapeutic antibodies. These antibodies primarily span IgG1, IgG2, and IgG4 isotypes. Experimentally derived developability metrics include: HIC retention time, SEC %Monomer, SMAC (surface membrane affinity chromatography), HAC (hydrophobic affinity chromatography), polyreactivity against CHO host cell proteins (PR_CHO) and ovalbumin (PR_Ova), AC-SINS at pH 6.0 and 7.4, melting temperatures T_{m1} and T_{m2}, and expression Titer.

Task A preprocessing (baseline developability metric and approval prediction): We filtered the PROPHET-Ab dataset down to 198 antibodies with binary approval labels (106 approved, 92 not approved). Missing biophysical values were imputed with the per-feature median. Table 1 summarizes missing value counts across the 11 selected features for all antibodies in PROPHET-Ab. HAC had the highest missingness (129 of 198 antibodies, approximately 65%), followed by PR_CHO, PR_Ova, and T_{m2} (41 each, approximately 21%), motivating the use of median imputation.

Table 1: Missing value counts for the 11 selected biophysical features across 198 labeled antibodies used in Task A approval prediction.

Feature	Missing	Feature	Missing
HIC	3	AC-SINS pH 6.0	3
SMAC	3	AC-SINS pH 7.4	3
HAC	129	Tm1	9
PR_CHO	41	Tm2	41
PR_Ova	41	Titer	4
SEC %Monomer	3		

Task B preprocessing (generative modeling): For the generative pipeline, fold 1 (50 antibodies) was held out for testing, and folds 2–5 (196 antibodies) were used for training, following the GDPa1 dataset's provided hierarchical-cluster, IgG isotype-stratified 5-fold splits. Sequences for the VH domain (average 110-130 residues) and VL domain (average 100-120 residues) were available for all 246 antibodies. HCDR3 lengths range from 5 to 24 residues (median 13), with no significant distribution shift between the training and test folds. The missing values in HIC and AC-SINS were imputed with the training median. Three types of embedding were computed per antibody: CDR3 region, framework region (excluding CDR3 region), and full sequence.

3.2 Observed Antibody Space (OAS) (Planned)

For pre-training the flow matching model, we plan to use paired VH/VL sequences from the OAS database, which contains over 2 billion antibody sequences from over 80 studies [6]. The paired subset contains millions of human antibody sequences. OAS pretraining is motivated by the framework-dominance challenge described in the challenges, risks, and mitigations section.

4 Baseline Implementation (Required)

4.1 Approval Prediction Baseline: XGBoost on Experimental Features

Baseline Model: An XGBoost binary classifier [2] was trained on 11 experimentally measured biophysical features from GDPa1: HIC, SMAC, HAC, PR_Ova, PR_CHO, SEC %Monomer, AC-SINS_pH6.0, AC-SINS_pH7.4, T_{m1}, T_{m2}, and Expression Titer. The model was configured with max_depth = 25, learning_rate = 0.25, n_estimators = 200, subsample = 0.8, colsample_bytree = 0.8, reg_alpha = 0.1, reg_lambda = 0.1, and objective = binary:logistic. This configuration matches the XGBoost hyperparameters reported in Arsiwala et al. [3]. Features were standardized using StandardScaler fitted on the training split only. Missing feature values were imputed with the per-feature median.

Training and Evaluation: Twenty independent XGBoost models were trained on random stratified 80:20 train-test splits. Each model produced a predicted approval probability for its test antibodies. The per-antibody developability score is defined as the mean predicted probability averaged across all bootstrap runs in which that antibody appeared in a test set. A null model was evaluated identically using permuted labels (permutation seed matched to train-test split seeds). The primary evaluation metrics were AUPRC and AUROC (AUPRC favored due to class imbalance in GDPa1). The separation between approved and non-approved developability scores was assessed using a one-sided Mann-Whitney U test. Feature importance scores were computed by training a single model on the full dataset (using random_state = 42).

4.2 Developability Prediction Baseline: Ridge Regression on p-IgGen Embeddings

Baseline Model: For each developability metric, a Ridge regression model (scikit-learn default $\alpha = 1.0$, no hyperparameter tuning) was trained on mean-pooled p-IgGen embeddings. Embeddings were computed by concatenating the VH and VL protein sequences with special begin (''1'') and end (''2'') tokens. Spaces were inserted between each amino acid character in the sequence string, followed by tokenizing with the p-IgGen tokenizer. The default forward pass through the model in batches of 16 was run, followed by extracting the final hidden layer and mean-pooling across all token positions (including the padding tokens).

Training and Evaluation: Per-target dropping of NA values was applied so that each metric utilizes a subset of antibodies with non-missing labels. Two evaluation protocols were used: (i) a single random 80:20 train-test split (`random_state=42`), and (ii) a leave-one-fold-out cross-validation using the hierarchical_cluster_IgG_isotype_stratified_fold column provided in GDPa1. This column partitions antibodies into 5 folds balanced by sequence similarity cluster and IgG isotype. In the cross-validation protocol, the Ridge model is retrained from scratch on the four training folds for each held-out fold. The cross-validation predictions are aggregated across all folds and an overall Spearman ρ is computed between all concatenated true and predicted values.

4.3 Approval and Developability Metric Baseline Results

Approval prediction. Table 2 summarizes the bootstrapped XGBoost results. The model achieved a median AUPRC of 0.665 and median AUROC of 0.622 across 20 bootstrap runs, compared to the null model's median AUPRC of 0.593 and AUROC of 0.524. The mean developability score was 0.619 ± 0.285 for approved antibodies versus 0.477 ± 0.316 for non-approved antibodies (one-sided Mann-Whitney $p = 7.3 \times 10^{-4}$), consistent with the separation reported by Arsiwala et al. (0.60 vs. 0.49, $p = 3.5 \times 10^{-5}$) [3]. The null positive rate was 0.535. Feature importance analysis identified HAC, T_{m2}, T_{m1}, and PR_Ova as the four most informative features (Table 3). Each of these features exceeded the mean feature importance of 0.091.

Table 2: XGBoost approval prediction: bootstrapped performance over 20 stratified 80:20 splits. AUPRC is the primary metric. The null positive rate is 0.535.

Model	Median AUPRC	Median AUROC
XGBoost (experimental features)	0.665	0.622
Null (shuffled labels)	0.593	0.524

Table 3: XGBoost feature importances (mean decrease in impurity), single model trained on all 198 labeled antibodies. Mean importance = 0.091.

Feature	Importance	Feature	Importance
HAC	0.166	Titer	0.072
Tm2	0.128	PR_CHO	0.070
Tm1	0.106	SMAC	0.068
PR_Ova	0.102	AC-SINS pH 7.4	0.066
SEC %Monomer	0.083	AC-SINS pH 6.0	0.063
HIC	0.075		

Developability metric prediction. Table 4 reports Spearman ρ for Ridge regression on p-IgGen embeddings across 10 developability metrics. Cross-validation estimates are consistently lower than random-split estimates for most metrics, confirming that sequence similarity inflates random-split performance. Under stratified cross-validation, the strongest signals are for PR_Ova ($\rho = 0.549$), PR_CHO ($\rho = 0.496$), and AC-SINS_pH7.4 ($\rho = 0.394$). HAC achieves the highest cross-validation ρ (0.621) but must be interpreted cautiously given its small effective sample size ($n = 94$). HIC achieves $\rho = 0.335$. Notably, T_{m2} and SEC %Monomer are near zero or negative under cross-validation, suggesting that these properties are not reliably recoverable from the p-IgGen embeddings. These results motivate the proposed AbLang2 fine-tuning as a stronger sequence-based predictor for the final report.

Table 4: Ridge regression on p-IgGen embeddings: Spearman ρ for 10 developability metrics under random 80:20 split and 5-fold hierarchical cluster, IgG isotype-stratified cross-validation. N = antibodies with non-missing labels. Cross-validation ρ is the primary reported figure; rows sorted by cross-validation ρ.

Target	N	Random-split ρ	Cross-Validation ρ
HAC	94	0.329	0.621
PR_Ova	197	0.598	0.549
PR_CHO	197	0.438	0.496
AC-SINS_pH7.4	242	0.456	0.394
HIC	242	0.408	0.335
Titer	239	0.477	0.239
SMAC	242	0.250	0.207
Purity	242	-0.019	0.180
Tm2	193	-0.111	-0.084
SEC %Monomer	242	0.174	-0.101

4.4 Generative Pipeline Baseline Implementation

Model type: Two separate Ridge regression models ($\alpha = 0.1$) are used as oracles to predict HIC and AC-SINS from AbLang2 full-sequence embeddings. A FiLM-conditioned MLP velocity network (CDR3VelocityNetPLS), consisting of two hidden layers of size 128 with GELU activations, LayerNorm, and 20% dropout, is trained using a conditional flow matching objective in a 10-dimensional PLS-reduced HCDR3 embedding space.

Input representation: The oracle takes 480-dimensional mean-pooled full-sequence AbLang2 seqcoding embeddings (the paired heavy-light chain sequence encoding) as input, standardized using a StandardScaler fit on the training set only. The flow model takes three inputs: the current 10-dimensional PLS position, a 480-dimensional masked framework embedding (CDR3 replaced with the mask token `'*'` before the AbLang2 forward pass), and a 2-dimensional scaled developability target (HIC and AC-SINS z-scores). Since only 197 training antibodies are available, 5$\times$ Gaussian noise augmentation is applied to the PLS and framework embeddings ($\sigma = 0.05$ and $\sigma = 0.01$ respectively), expanding the training set to 1,182 examples.

Training setup: The oracle has a closed-form solution (no gradient-based training). The flow model is trained with MSE loss on the CFM velocity target ($\sigma = 0.5$), using AdamW (lr $= 10^{-3}$, weight decay $= 10^{-4}$), cosine annealing over 1,000 epochs (minimum lr $= 10^{-5}$), batch size 32, and gradient clipping at max norm 1.0. No early stopping is used. Training was performed on a T4 GPU.

Evaluation protocol: The oracle is evaluated on the held-out fold-1 test set ($n = 50$) using Spearman ρ, consistent with prior developability prediction literature [3]. To verify the oracle does not overfit, 4-fold cross-validation is performed within the training folds (folds 2–5), with a separate StandardScaler refit inside each inner loop. The flow model is evaluated end-to-end through the three-phase generation pipeline: for each of the 50 test antibodies, $M = 20$ CDR3 sequences are generated per condition and the oracle selects the best by composite HIC + AC-SINS z-score. Performance is compared across parent, baseline (unguided AbLang2, guidance_strength $= 0$), and guided (flow-steered, guidance_strength $= 2.0$) conditions using paired t-tests. Two additional metrics not used during training (CamSol solubility and sequence SAP score) are reported as independent validation of biophysical improvement.

4.5 Generative Pipeline Baseline Results

Oracle performance: The Ridge regression oracle trained on AbLang2 full-sequence embeddings achieves Spearman $\rho = 0.32$ for HIC and $\rho = 0.445$ for AC-SINS on the held-out fold-1 test set. The oracle is trained on full-sequence embeddings because HIC and AC-SINS are whole-molecule properties.

Biophysical property evaluation of generated sequences: Table 5 compares three conditions across $n = 50$ held-out fold-1 antibodies: the parent (unmodified) sequence, the AbLang2-only baseline (unguided), and the flow-guided generation. Four metrics are reported: oracle-predicted

HIC and AC-SINS (the oracle's scoring targets), CamSol solubility (computed analytically from the generated CDR3 sequence, not used in training), and SAP score (hydrophobic patch exposure, not used in training).

Table 5: Biophysical evaluation of generative pipeline across $n = 50$ held-out test antibodies (mean $\pm$ SE). Arrows indicate direction of improvement. CamSol and SAP are independent of oracle training.

Metric	Parent	Baseline (AbLang2)	Guided (Flow + AbLang2)
Oracle HIC (↓)	2.738 ± 0.048	2.466 ± 0.043	2.493 ± 0.046
Oracle AC-SINS (↓)	3.741 ± 1.215	-0.545 ± 1.178	-0.441 ± 1.044
CamSol (↑)	-0.004 ± 0.056	0.564 ± 0.071	0.649 ± 0.078
SAP (↓)	0.449 ± 0.016	0.216 ± 0.019	0.242 ± 0.020

Paired t-tests reveal that both generation conditions achieve statistically significant improvements over the parent on all four metrics ($p < 0.001$ in each case; Table 6). Crucially, guided and baseline conditions perform comparably on oracle metrics (HIC: $\Delta_{\text{guided}} = -0.245$ vs. $\Delta_{\text{baseline}} = -0.271$; AC-SINS: $\Delta_{\text{guided}} = -4.18$ vs. $\Delta_{\text{baseline}} = -4.29$), suggesting that the flow model's explicit conditioning contributes marginal additional steering at this data scale.

Table 6: Paired t-tests comparing generation conditions vs. parent ($n = 50$). Δ = mean change relative to parent. All tests one-sided. * denotes $p < 0.001$.

	Baseline vs. Parent		Guided vs. Parent	
Metric	Δ	p	Δ	p
Oracle HIC	-0.271	$< 0.001^*$	-0.245	$< 0.001^*$
Oracle AC-SINS	-4.286	$< 0.001^*$	-4.182	$< 0.001^*$
CamSol	$+0.568$	$< 0.001^*$	$+0.653$	$< 0.001^*$
SAP	-0.233	$< 0.001^*$	-0.207	$< 0.001^*$

The substantial improvement on CamSol and SAP, which are not oracle training targets, warrants further investigation. However, since unguided AbLang2 generation alone produces comparable gains, these improvements likely reflect AbLang2's language model prior favoring less hydrophobic CDR3 sequences rather than the flow model's targeted optimization. This is the primary finding and a key limitation discussed in the challenges and mitgations section.

5 Proposed Method: Design and Implementation Status

Our approach consists of two main components: sequence-based prediction (Task A) and generative optimization (Task B).

Task A: Sequence-Based Prediction For the main model, we will fine-tune the AbLang2 encoder [5] on the GDPa1 dataset. The embeddings from fine-tuning will be passed to a lightweight prediction head (a linear classifier for approval or a Ridge regression model for the 10 biophysical metrics evaluated in the baseline model with p-IgGen). In terms of parts that are still under development, fine-tuning of AbLang2 with LoRA [13] will constitute the primary Task A work for the second half of the project.

Task B: Generative Developability Optimization The generative pipeline consists of three interacting components. First, a Ridge regression oracle is trained on full-sequence AbLang2 embeddings to predict HIC and AC-SINS; this oracle scores candidate sequences during generation. Second, the CDR3 embedding from AbLang2 (480-dimensional) is compressed to a 10-dimensional PLS latent space by maximizing covariance with HIC and AC-SINS labels, ensuring that every latent dimension encodes developability-relevant variation. Third, a FiLM-conditioned [11] velocity network is trained using the conditional flow matching (CFM) objective to transport Gaussian noise to the PLS-CDR3 distribution. The network is conditioned on (i) the masked framework embedding (480-dimensional,

with CDR3 zeroed out), providing antibody context, and (ii) desired HIC and AC-SINS target values, steering generation toward improved developability.

At inference, for each test antibody: (1) the flow model generates 50 CDR3 targets in PLS space conditioned on the parent framework and 25th-percentile target values; (2) for each HCDR3 position, AbLang2 decoding is reweighted toward the top-5 PLS targets, producing 20 biased candidate sequences; (3) the oracle selects the best-scoring candidate. This guided procedure is compared against an unguided baseline that uses AbLang2 decoding alone with oracle-best selection.

Implementation status The full generative pipeline is implemented and has produced preliminary results. Ridge regression baselines on p-IgGen embeddings are fully evaluated across 10 developability metrics. Fine-tuning the AbLang2 sequence model with LoRA is currently under development.

5.1 Planned Experiments Beyond the Baseline

- **AbLang2 fine-tuning for Task A:** LoRA fine-tuned AbLang2 for approval classification and per-metric regression, compared against XGBoost (upper bound), and p-IgGen Ridge (sequence-only baseline).
- **Improved Oracle:** Evaluate whether fine-tuned AbLang2 embeddings improve developability regression compared to the p-IgGen baseline.
- **OAS pretraining for flow model:** Pretrain on OAS CDR3/framework pairs to reduce framework dominance, then fine-tune on PROPHET-Ab.
- **Sequence-level Plausibility Analysis:** Evaluate generated HCDR3 sequences for amino acid composition, germline proximity, and CDR3 length distribution relative to known therapeutic antibodies.
- **Binding Preservation:** Assess whether optimization reduces predicted binding affinity using AlphaFold-Multimer or potentially Rosetta-based interface scoring.

6 Challenges, Risks, and Mitigation

For each, describe what you have tried so far and your mitigation plan (e.g., simpler models, regularization strategies, subsampling, alternative datasets).

Data scarcity: PROPHET-Ab contains only 246 antibodies total, with substantially fewer per metric after missingness filtering (e.g., $n = 242$ for HIC, $n = 94$ for HAC). This could potentially limit model capacity and could increase cross-validation variance. Some of the mitigation strategies to address this limitation were applied when building the generative model such as PLS dimensionality reduction from 480d to 10d for the flow model, freezing AbLang2 as a backbone to avoid overfitting, and minor sequence perturbation augmentation during flow training.

Framework-embedding dominance: The masked framework embedding dominates the flow model's conditioning signal, generating CDR3 sequences that strongly resemble the parent regardless of the developability conditioning target values. PLS space visualization confirms that flow model targets cluster around the parent CDR3 embedding irrespective of the specified HIC/AC-SINS targets, explaining the lack of guided-vs-baseline improvement. One way to mitigate this would be to pre-train with OAS in order to expose the flow model to a wider CDR3 distribution before fine-tuning. In addition, ablation experiments to assess the contribution of each conditioning input could clarify which inputs drive changes in predicted developability metrics.

Oracle reliability: The HIC/AC-SINS oracle (Spearman $\rho = 0.335$ for HIC under cross-validation) has limited accuracy, meaning optimization against it may not reflect true biophysical improvement. Some of the mitigation strategies for this challenge include reporting held-out CamSol and SAP metrics not used in training, and developing AbLang2 fine-tuned oracles with better predictive performance.

7 Updated Timeline and Plan to Final Report

- **Weeks 1–2:** Complete AbLang2 fine-tuning pipeline for Task A (approval prediction and per-metric regression). Structural/binding analysis of generated sequences.
- **Week 3:** Begin OAS sequence preprocessing for flow model pretraining. Fine-tune OAS-pretrained flow model on PROPHET-Ab.
- **Week 4:** Sequence diversity and plausibility analysis. Rerun guided generation experiments. Hyperparameter sweep for generation. Draft final report.
- **Week 5:** Finalize figures, tables, and writing. Revise discussion in light of all results. Submit final report.

8 Appendix: Antibody Developability Definitions and Metrics

Developability metrics refer to the biophysical properties of an antibody that must be considered during lead optimization and development [1].

Thermal stability ($\mathbf{T_{m1}}, \mathbf{T_{m2}}$): Measures the melting temperature of antibody domains. Low T_m values indicate susceptibility to poor manufacturing yields, low solubility, and increased aggregation. T_{m1} refers to the melting temperature of the CH_2 domain (i.e., representing the lowest thermal transition). T_{m2} corresponds to the temperature at which the Fab domain unfolds. The T_{m2} metric is particularly informative for therapeutic antibodies and has been identified as a strong predictor of approval status [1][3].

Hydrophobicity and non-specific surface binding (HIC, SMAC, HAC, SAP score): Describes the tendency of surface-exposed hydrophobic patches to influence non-specific binding, membrane interactions, and aggregation propensity. HIC chromatography and SMAC (standup monolayer affinity chromatography, which measures binding to a lipid membrane surface) both quantify hydrophobic non-specific binding. HAC (heparin affinity chromatography) measures binding to a negatively charged surface and reflects an excess positive surface charge. This metric is also associated with polyreactivity and rapid clearance (premature removal of a therapeutic antibody candidate from circulation). The spatial aggregation propensity (SAP) score computes residue-level hydrophobic exposure from structure. Increased hydrophobicity is associated with shorter half-lives of antibodies [1][3].

Self-association and polyreactivity (AC-SINS, PR_CHO, PR_Ova): Captures the propensity of an antibody to bind non-specifically to itself or to common biological surfaces. AC-SINS (affinity capture self-interaction nanoparticle spectroscopy) measures self-association, while PR_CHO and PR_Ova quantify polyreactivity through binding to CHO cell proteins and ovalbumin respectively. Increased self-association metrics and polyreactivity are linked with elevated antibody self-interaction (often causing high viscosity or gelation in high-concentration formulations) as well as a greater tendency toward non-specific multi-antigen binding [1][3][16].

Colloidal Stability (SEC %Monomer, Purity): Measures the portion of the antibody that stays in its native monomeric state. Maintaining high monomeric purity is essential for developability, since aggregated species can trigger adverse immune responses and diminish the given therapeutic antibody's effective dose [1][3].

Expression Titer: Reflects the concentration/expression yield of antibody produced in cell culture (typically CHO cells). This metric serves as a key determinant of manufacturing efficiency and cost [3]. Low expression titers often indicate issues with protein folding, stability, or secretion. This can render candidates infeasible for production. On the other hand, higher expression titers can enable scalable and cost-effective production [17].

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Predictive and Generative Modeling of Therapeutic Antibody Developability Using Fine-Tuned Protein Language Models

Sidhant Puntambekar & Jack Hwang

BMI-702 Midterm Presentation

Background and Motivation

- Antibodies are Y-shaped proteins built from two chains:
  - **VH (Heavy-chain region)**: Contributes to antigen binding
  - **VL (Light-chain region)**: Pairs with VH, forms complete binding site

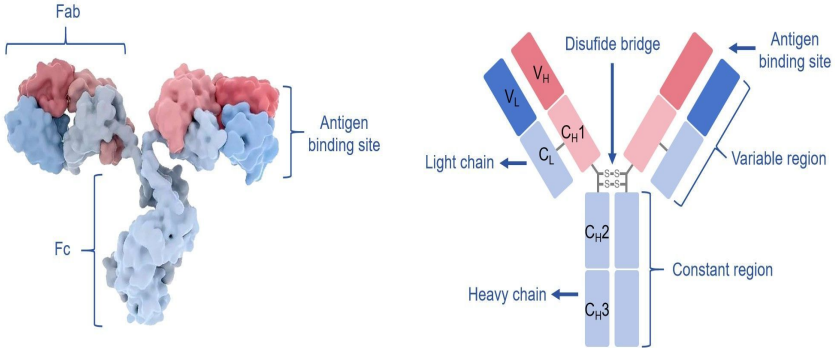
- Therapeutic antibodies are widely used but expensive and slow to develop

  ↳ Many candidates fail due to poor developability

  - Developability metrics include biophysical properties:
    - Thermostability (T_m)
    - Self-association propensity (AC-SINS)
    - Hydrophobicity (HIC, SAP)
    - Solubility

- Lead optimization focuses on modifying antibody sequences to improve these properties

  ↳ Requires balancing developability while preserving antibody binding function



Source: Creative Biolabs Antibody Development

Research Question: From Sequence to Developability

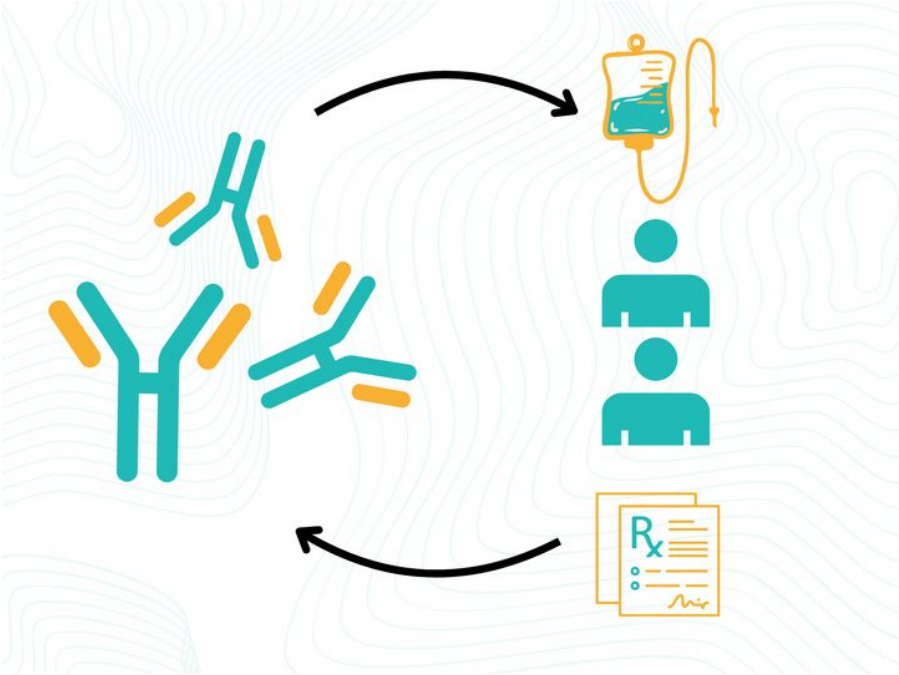
Main Research Question:

Can sequence-based protein language models learn representations that:

1. Predict developability properties and therapeutic approval?
2. Enable optimization of true developability properties through guided sequence generation?

- Prior work used experimentally measured features from antibody sequences to predict developability and downstream approval classification (Arsiwala et al., 2025)

- However, experimental features are not available early
  ↳Motivates the need for sequence-only ML models to predict developability



Source: Georgia Channing

Project Overview: Predict and Optimize Antibodies

Task A: Antibody Sequence Scoring

Input: Antibody sequence, VL and VH amino acid sequences

Output: Developability metrics and binary approval likelihood

Baseline Model: Ridge regression using p-IgGen for developability metrics, XGBoost for approval likelihood

Model: AbLang2 encoder -> embedding, with a classifier head (fine-tuned with LoRA)

Training Data: Ginkgo Bioworks PROPHET-Ab (GDPa1_v1.2) — 197 therapeutic antibodies

Task B: Antibody Sequence Generation

Input: Lead antibody CDR3, framework embedding, and desired HIC and AC-SINS target scores

Output: Generated CDR3 variants of the same length as the parent, with improved predicted developability (lower HIC, lower AC-SINS)

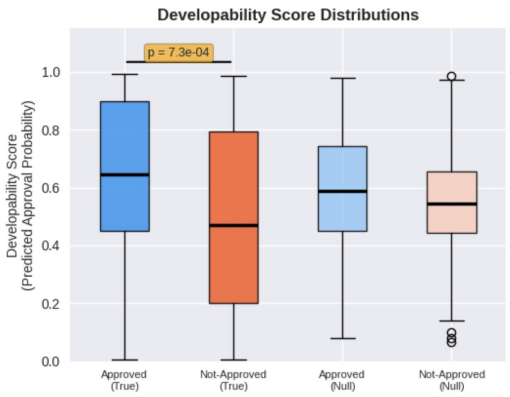
Model: Conditional flow matching in AbLang2 CDR3 PLS-reduced embedding space (480d → 10d), with FiLM conditioning on masked framework embeddings; AbLang2 MLM used as token-level sequence decoder, steered by flow model targets

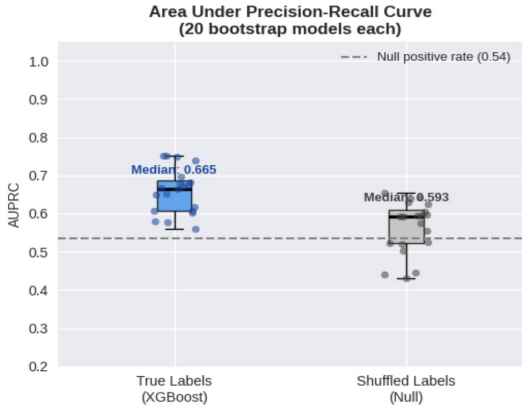
Training Data: Ginkgo Bioworks PROPHET-Ab (GDPa1_v1.2) — 197 therapeutic antibodies (folds 2–5)

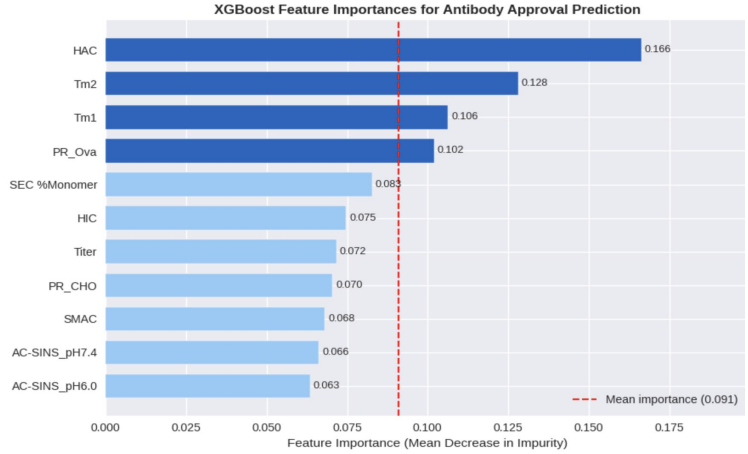
Therapeutic Approval Prediction Baseline Modelling

- **Data**: GDPa1 dataset with 197 antibodies, 30 raw feature columns
  - **Feature Selection**: 11 developability features (HIC, SMAC, HAC, PR_Ova, PR_CHO, SEC %Monomer, AC-SINS_pH6.0, AC-SINS_pH7.4, Tm1, Tm2, Titer)
  - **Binary classification**: 'Approved' (106 antibodies) vs. 'Not-Approved' (92 antibodies)
  - **Developability Score:** Per-sample mean predicted probability from XGBoost

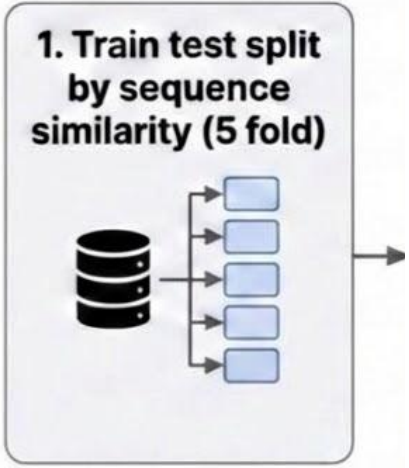
- Bootstrapped twenty independent XGBoost models trained on random 80:20 stratified train-test splits
  - Uses 11 experimentally measured biophysical features
  - Feature importances calculated by training a single model on all data

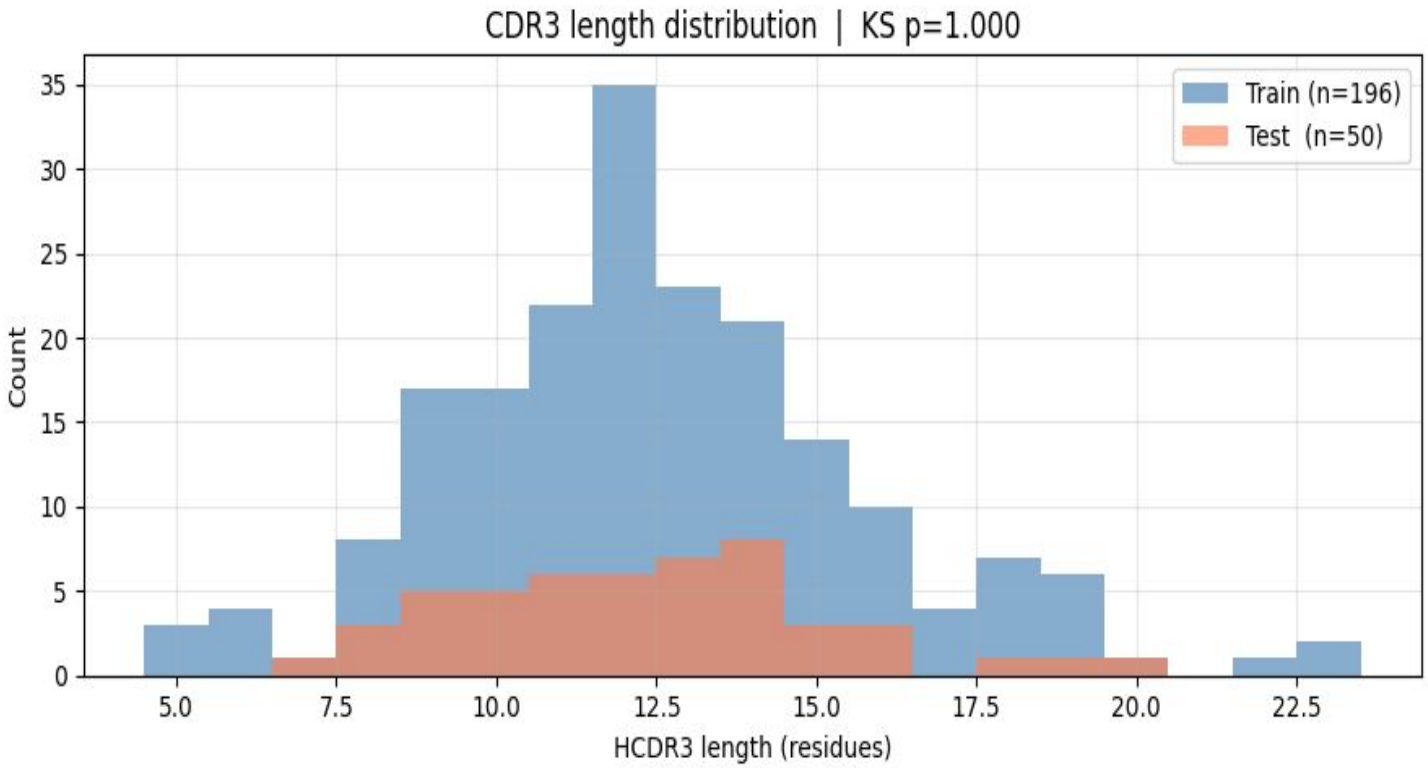






Generative Model Training Pipeline

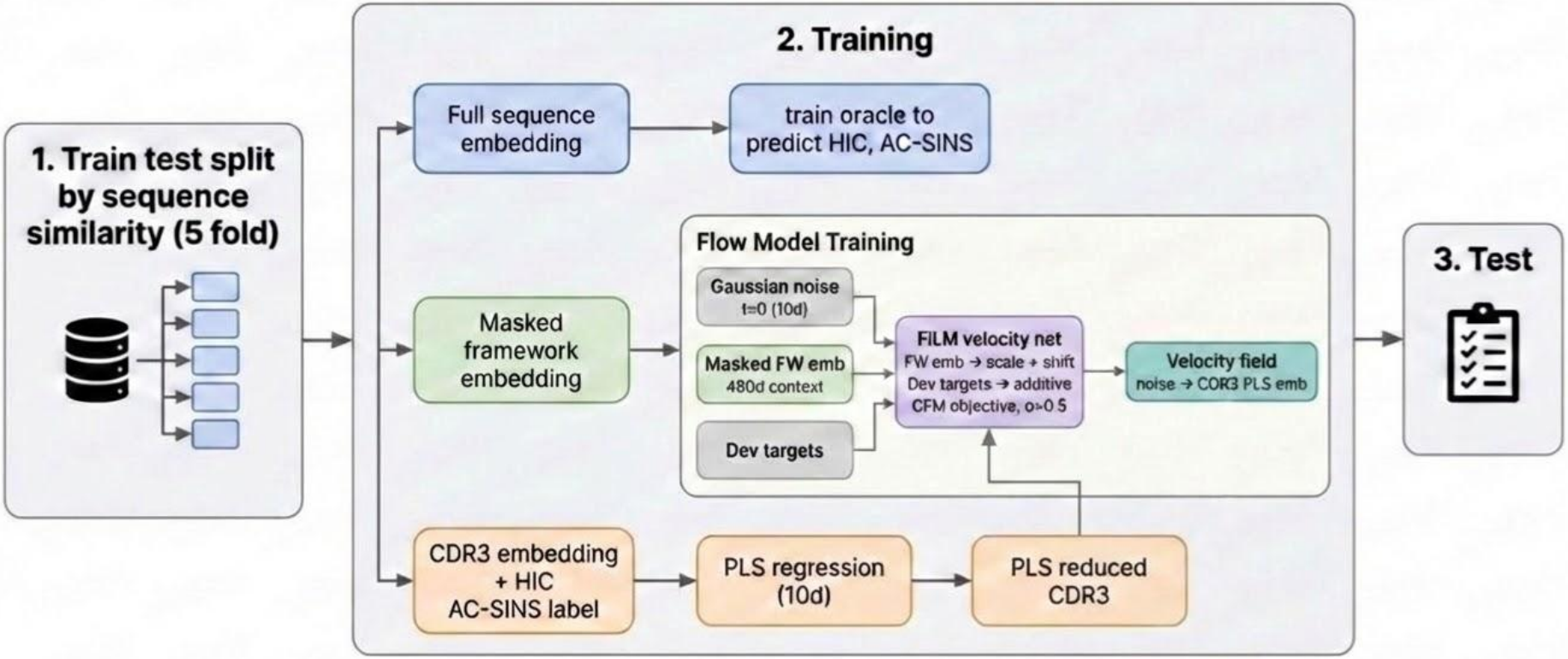




PLS: maximises covariance with HIC and AC-SINS (every dimension is developability relevant)

AC-SINS: Self-association
HIC: Hydrophobicity

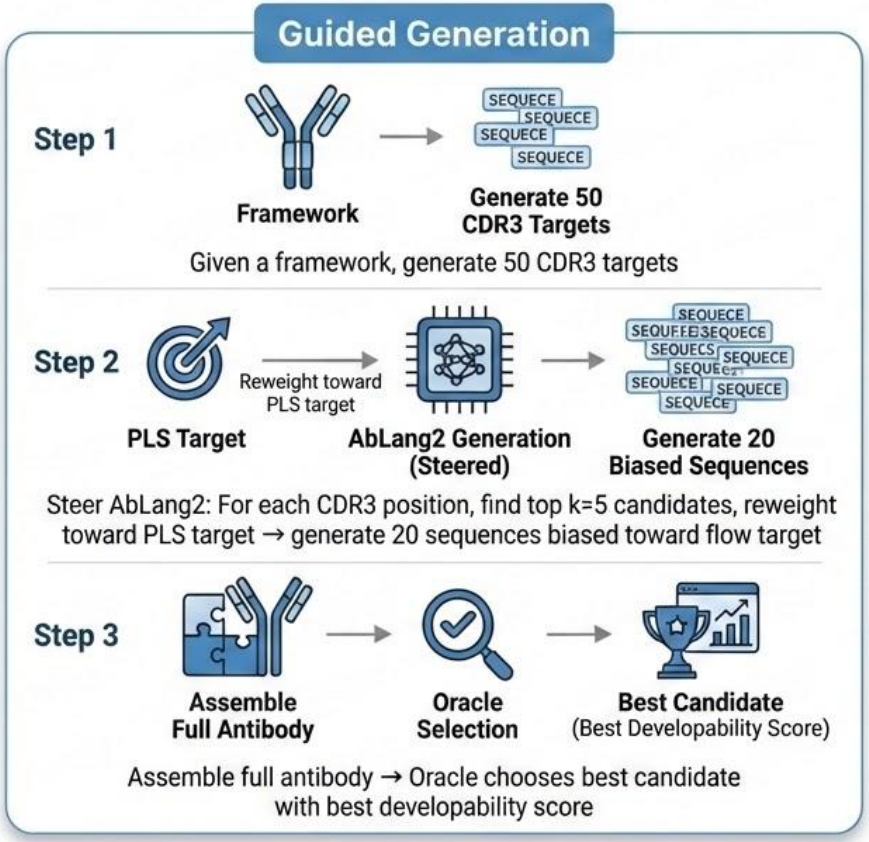
Generative Model Training Pipeline

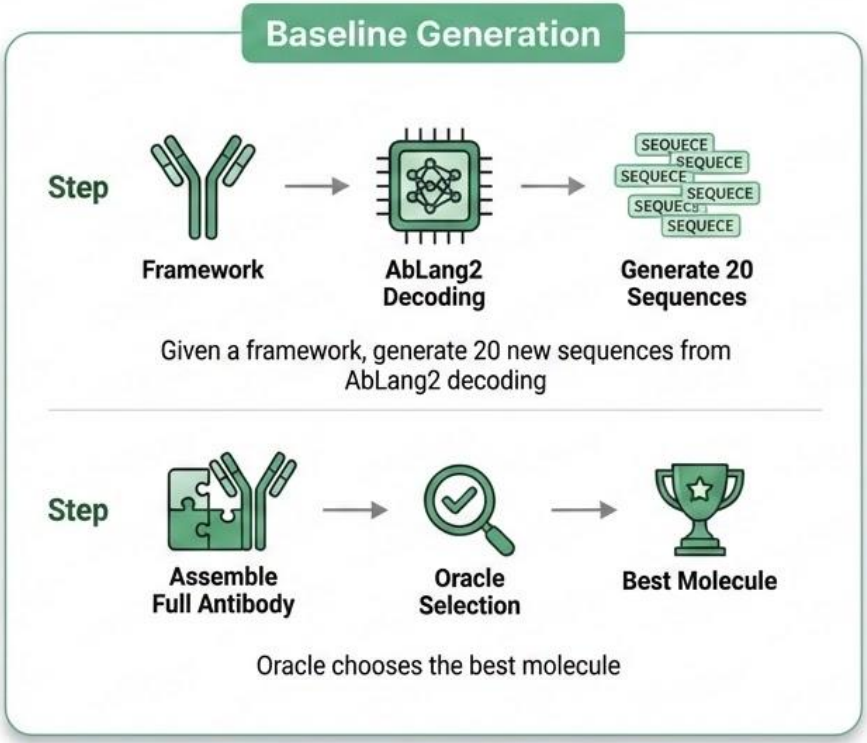


PLS: maximises covariance with HIC and AC-SINS (every dimension is developability relevant)

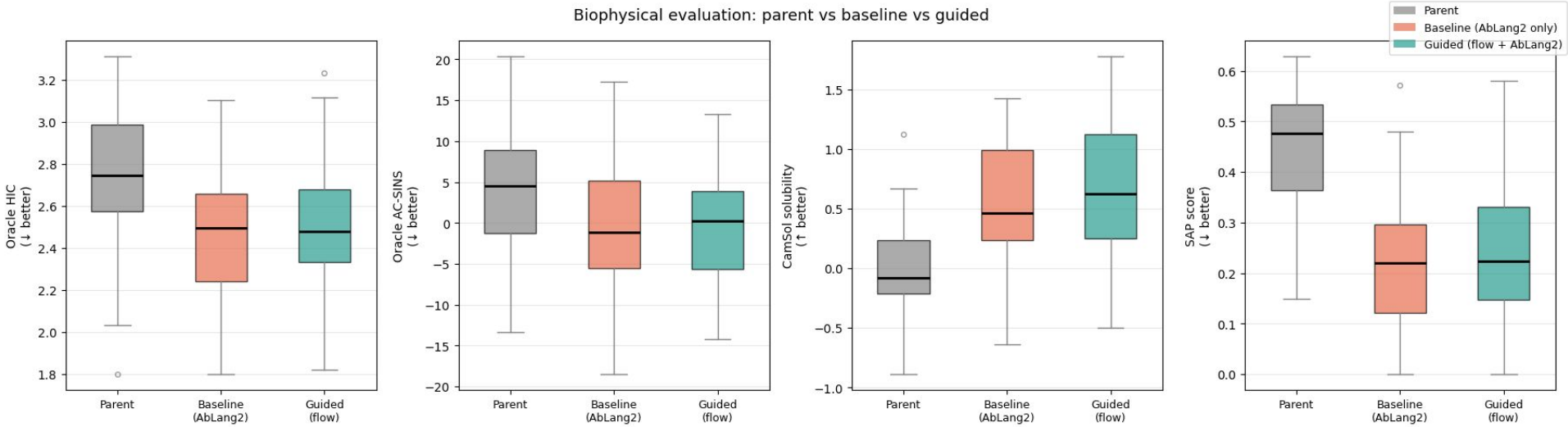
AC-SINS: Self-association
HIC: Hydrophobicity

Generation - Guided with flow and Baseline (only AbLang2)





Result: AbLang2 Performs Well & Guided Generation generates favorable Abs



Metric	Parent	Baseline	Guided
Oracle HIC (↓)	+2.738±0.048	+2.466±0.043	+2.493±0.046
Oracle SINS (↓)	+3.741±1.215	−0.545±1.178	−0.441±1.044
CamSol (↑)	−0.004±0.056	+0.564±0.071	+0.649±0.078
SAP (↓)	+0.449±0.016	+0.216±0.019	+0.242±0.020

Paired t-tests (* = p < 0.05):

Metric	Baseline vs Parent	Guided vs Parent	Guided vs Baseline
Oracle HIC (↓)	Δ=−0.271 p=0.000*	Δ=−0.245 p=0.000*	Δ=+0.027 p=0.418
Oracle SINS (↓)	Δ=−4.286 p=0.000*	Δ=−4.182 p=0.000*	Δ=+0.104 p=0.872
CamSol (↑)	Δ=+0.568 p=0.000*	Δ=+0.653 p=0.000*	Δ=+0.085 p=0.463
SAP (↓)	Δ=−0.233 p=0.000*	Δ=−0.207 p=0.000*	Δ=+0.026 p=0.414

For ↓ metrics: Δ < 0 = improvement.
For ↑ metrics: Δ > 0 = improvement.

AC-SINS: Self-association
HIC: Hydrophobicity
CamSol: Computed solubility
SAP: Aggregation due to hydrophobic patches

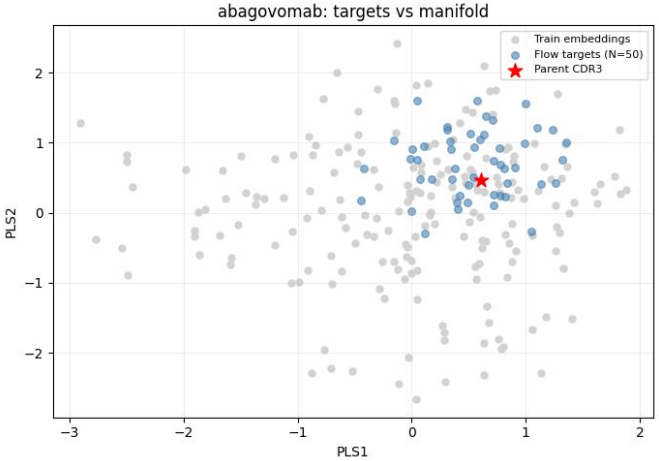
Challenges, Limitations & Solutions

1. Data scarcity

⇒ Had to use PLS reduction, frozen AbLang2, data augmentation

2. The framework conditioning seems to dominate the embedding generation

⇒ Pretrain the flow model on OAS CDR3/framework pairs and fine tune on Prophet-Ab



PLS1: HIC ρ=+0.291 SINS ρ=−0.570
PLS2: HIC ρ=−0.374 SINS ρ=−0.326
PLS3: HIC ρ=−0.388 SINS ρ=−0.100

Next Steps - Generative Antibody Modelling

- Tune generative optimization pipeline and analysis of the embedding space
  - Tune key parameters (number of candidates, temperature for AbLang2 decoding) to balance improvement and diversity
  - Check if similar substructures appear in similar embedding space

- Add additional developability scores such as chemical liabilities, $\Delta\Delta G$

- **Conditioning the generation on Antigen binding affinity and specificity**
  - Measure the binding affinity and specificity of target antigen with the optimized molecule to ensure the lead optimization doesn't reduce the binding affinity and specificity

Next Steps - Developability and Approval Prediction

- Train and evaluate sequence-based prediction model
  - Fine-tune AbLang2 on PROPHET-Ab to predict developability metrics from sequence
    - There are room for the current oracles performance to improve → Fine Tuning could help
  - Compare performance against XGBoost to assess how much developability scores can be learned without experimentally derived features

- Build and validate developability predictors
  - Evaluate correlation with experimental measurements to ensure reliable optimization signals

Singular Ask For Feedback:

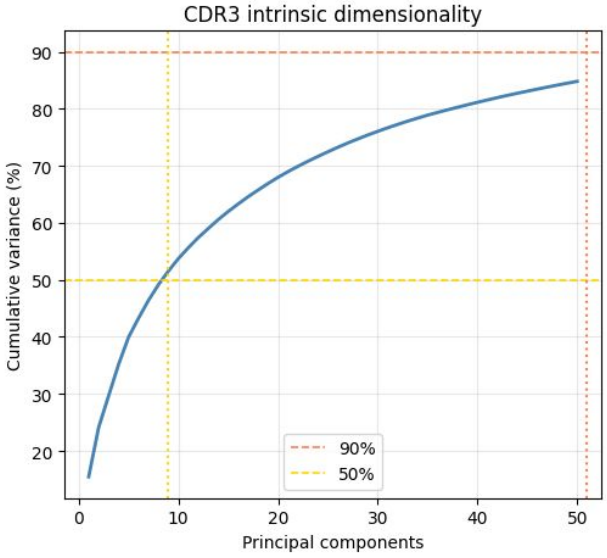
Are the biophysical properties (T_m , HIC, AC-SINS, solubility) sufficient to guide meaningful improvements in antibody developability, or should we predict additional properties to better capture real-world developability constraints?

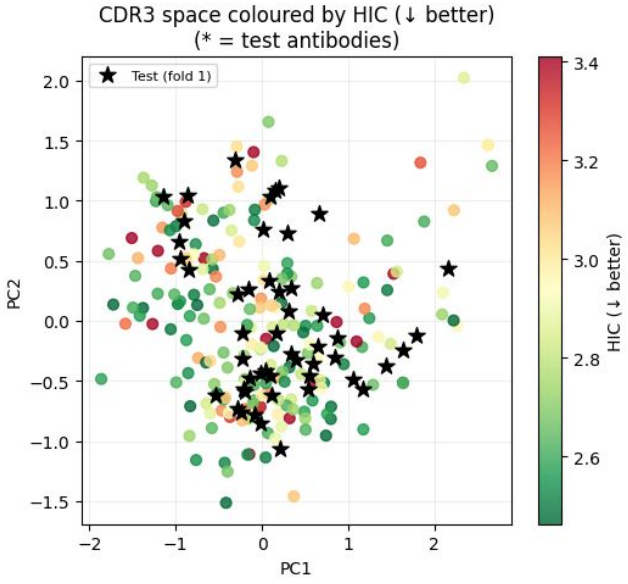
Roles

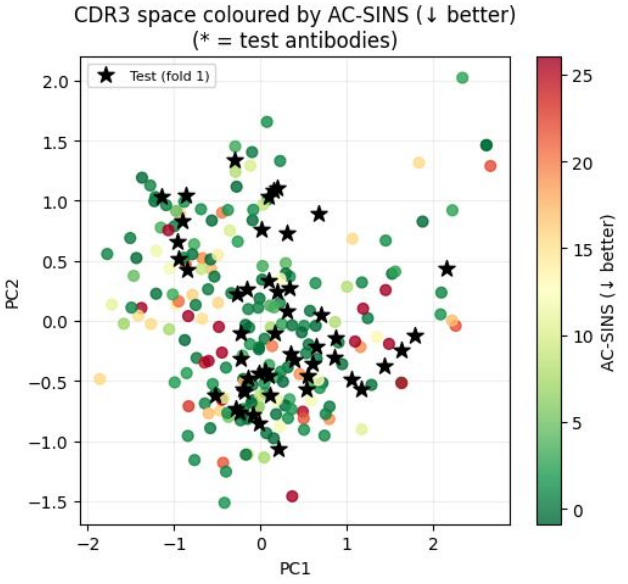
Sidhant: Built baseline models for developability metrics (ridge regressions with p-IgGen) and antibody approval (XGBoost), building fine tuned sequence based model (using AbLang2) for developability metric prediction and antibody approval

Jack: Built generative pipeline for developability optimization (dimensionality reduction of CDR3 embeddings using PLS, training a conditional flow matching model in reduced embedding space, and implementing a steered sequence generation procedure using AbLang2 guided by flow model targets)

Appendix 1


CDR3 intrinsic dimensionality
Cumulative variance (%)
Principal components
90%
50%


CDR3 space coloured by HIC (↓ better)
(* = test antibodies)
Test (fold 1)
PC1
PC2
HIC (↓ better)


CDR3 space coloured by AC-SINS (↓ better)
(* = test antibodies)
Test (fold 1)
PC1
PC2
AC-SINS (↓ better)